

The Toric Code

Prachi Tyagi

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1 Lattice Structure

The Toric Code is a topological quantum error-correcting code defined on a two-dimensional periodic lattice.

For an $L \times L$ square lattice, physical qubits are placed on the edges of the lattice. The total number of physical qubits is

$$n = 2L^2.$$

The following figure shows an $L \times L$ lattice for $L = 5$, with physical qubits placed on all edges.

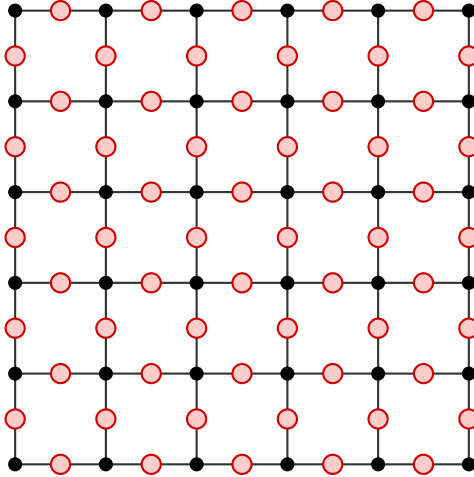


Figure 1: $L \times L$ (5×5) Toric Code lattice with physical qubits (red circles) placed on all edges.

The lattice has periodic boundary conditions. The opposite boundaries are identified:

- The top boundary is connected to the bottom boundary.
- The left boundary is connected to the right boundary.

Therefore, the lattice has the topology of a torus.

2 Stabilizers

The Toric Code is a CSS stabilizer code with two types of stabilizer operators: star operators and plaquette operators.

There are L^2 vertices and L^2 plaquettes.

2.1 Star Operators

For every vertex v , the star operator is defined as

$$A_v = \prod_{e \ni v} X_e,$$

where the product is taken over the four edges incident on vertex v .

Therefore, each star operator has weight

$$\text{wt}(A_v) = 4.$$

2.2 Plaquette Operators

For every plaquette p , the plaquette operator is defined as

$$B_p = \prod_{e \in \partial p} Z_e,$$

where the product is taken over the four edges forming the boundary of the plaquette. Thus,

$$\text{wt}(B_p) = 4.$$

The stabilizer group is generated by all star and plaquette operators:

$$S = \langle A_v, B_p \rangle.$$

3 Independence of Stabilizer Generators

Although there are L^2 star operators and L^2 plaquette operators, they are not all independent.

Since every edge belongs to exactly two stars,

$$\prod_v A_v = I.$$

Similarly, since every edge belongs to exactly two plaquettes,

$$\prod_p B_p = I. \tag{1}$$

Therefore,

$$\text{independent } X\text{-type generators} = L^2 - 1,$$

and

$$\text{independent } Z\text{-type generators} = L^2 - 1.$$

Hence, the stabilizer rank is

$$r = 2(L^2 - 1).$$

Since

$$k = n - r,$$

we obtain

$$\begin{aligned}
k &= 2L^2 - 2(L^2 - 1) & (2) \\
&= 2. & (3)
\end{aligned}$$

Thus, the Toric Code encodes two logical qubits.

4 Code Parameters

The Toric Code therefore has parameters

$$\boxed{[[2L^2, 2, L]]}.$$

Here,

- $n = 2L^2$ is the number of physical qubits,
- $k = 2$ is the number of logical qubits,
- $d = L$ is the code distance.

5 Logical Operators

The logical Pauli operators can be represented by non-contractible loops on the torus. We denote the logical operators by

$$\bar{X}_1, \quad \bar{X}_2, \quad \bar{Z}_1, \quad \bar{Z}_2.$$

The logical operators satisfy

$$\bar{X}_i \bar{Z}_i = -\bar{Z}_i \bar{X}_i, \quad i = 1, 2.$$

Logical operators corresponding to different logical qubits commute:

$$[\bar{X}_1, \bar{Z}_2] = 0,$$

$$[\bar{X}_2, \bar{Z}_1] = 0.$$

6 Trivial and Non-Trivial Loops

Errors in the Toric Code can be visualized as chains of Pauli operators acting on lattice edges.

6.1 Trivial Loops

A contractible loop can be continuously shrunk to a point.

For example, the boundary of a single plaquette is

$$B_p = \prod_{e \in \partial p} Z_e.$$

Larger contractible loops can be written as products of plaquette stabilizers.

Therefore,

$$\text{contractible loop} \in S.$$

Such an operator acts trivially on the logical information.

6.2 Non-Contractible Loops

A non-contractible loop cannot be continuously shrunk to a point.

It winds around one of the two periodic directions of the torus.

There are two independent winding directions:

- horizontal,
- vertical.

These non-contractible loops correspond to logical operators.

A non-contractible Z -string can represent a logical $\bar{Z}$ operator, while an appropriate non-contractible X -string represents its conjugate logical $\bar{X}$ operator.

7 Minimum Distance

The distance of a quantum stabilizer code is the minimum weight of a non-trivial logical operator:

$$d = \min_{P \in N(S) \setminus S} \text{wt}(P).$$

For the Toric Code, the smallest non-contractible loop must wind completely around the torus.

For an $L \times L$ lattice, such a loop contains at least L edges. Therefore,

$$\boxed{d = L}.$$

Thus,

$$\boxed{[[2L^2, 2, L]]}. \tag{4}$$

8 Errors and Syndromes

Suppose a Pauli error E acts on the physical qubits.

For a stabilizer S_i , a code state satisfies

$$S_i|\psi\rangle = |\psi\rangle.$$

If E commutes with S_i ,

$$S_i E|\psi\rangle = E|\psi\rangle,$$

and the measurement outcome remains $+1$.

If E anticommutes with S_i ,

$$S_i E|\psi\rangle = -E|\psi\rangle, \tag{5}$$

and the measurement outcome becomes -1 .

Therefore,

- $+1$: no syndrome defect,
- -1 : syndrome defect.

The collection of all -1 stabilizer outcomes is called the **syndrome**.

9 Error Chains and Syndrome Defects

Errors in the Toric Code form chains on the lattice edges.

For example, a chain of Z errors can produce syndrome defects at the endpoints of the chain.

Therefore,

error chain $\longrightarrow$ syndrome defects at its endpoints.

A closed error chain has no endpoints and therefore produces no syndrome defects.

There are three important cases:

- Open chain: produces syndrome defects.
- Closed contractible loop: corresponds to a stabilizer.
- Closed non-contractible loop: corresponds to a logical operator.

10 Syndrome Example

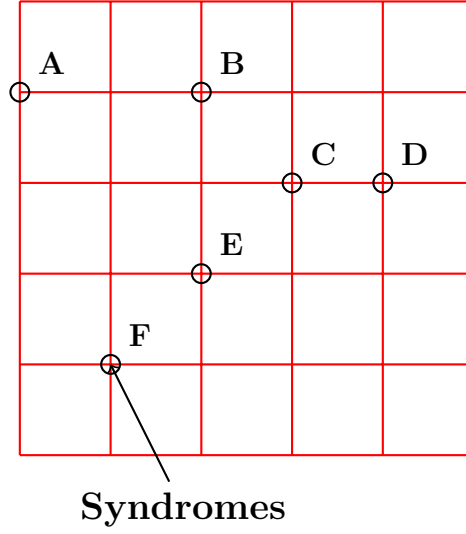


Figure 2: Toric Code lattice showing syndrome defects A, B, C, D, E, and F.

Consider a syndrome pattern containing defects at locations A, B, C, D, E, F .

These locations represent stabilizers whose measurement outcome is -1 .

The syndrome does not, in general, uniquely determine the original error pattern.

Different error chains can produce the same syndrome.

Therefore, the decoder must infer a likely error chain consistent with the observed syndrome.

Syndrome $\longrightarrow$ Likely Error Chain

11 Minimum-Weight Perfect Matching

The Toric Code can be decoded using a **Minimum-Weight Perfect Matching (MWPM)** decoder.

The decoding procedure is:

1. Measure all stabilizers.
2. Identify the syndrome defects.
3. Calculate distances between syndrome defects.
4. Pair the defects using MWPM.
5. Construct correction chains between the matched defects.
6. Apply the corresponding corrections.
7. Check whether a logical error remains.

The decoder attempts to find a correction chain with minimum total weight.

12 Physical Errors and Logical Errors

A physical error does not necessarily corrupt the encoded logical information.

If an error is equivalent to a stabilizer,

$$E \in S,$$

it acts trivially on the codespace.

However, if the residual error corresponds to a non-trivial logical operator,

$$E \in N(S) \setminus S,$$

then the logical state is changed.

Thus,

$$\boxed{\text{physical error} \neq \text{necessarily logical error}}.$$

13 Summary

For an $L \times L$ Toric Code,

$$n = 2L^2, \quad k = 2, \quad d = L.$$

Therefore,

$$\boxed{[[2L^2, 2, L]]}.$$

The complete error-correction process can be summarized as

$$\boxed{\text{Physical Errors} \rightarrow \text{Syndrome} \rightarrow \text{MWPM Decoding} \rightarrow \text{Correction} \rightarrow \text{Logical Error Check}}$$